B25ADT303 Project work Phase I

FIRST REVIEW

Department of Computer Science and Engineering KIT –
Kalaignarkarunanidhi Institute of Technology Academic Year 2026–2027

Project Title - Intelligent Street Lighting Control System

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Introduction

Project Background and Context

- Street lighting is an essential part of modern urban infrastructure, providing visibility and safety for vehicles and pedestrians.
- Conventional street lights commonly operate using fixed schedules or fixed brightness levels.
- During low-traffic periods, maintaining high illumination can result in unnecessary electricity consumption.
- Manual monitoring and control of large numbers of street lights is difficult and time-consuming.
- Smart-city infrastructure requires automated systems that can respond to real-time environmental and traffic conditions.
- Artificial Intelligence, Computer Vision and IoT can be combined to enable adaptive and intelligent street-light control.

Introduction

Problem Area and Motivation

Existing Problem

- Fixed-brightness street lights consume power even when road activity is low.
- Timer-based systems cannot understand real-time traffic density.
- Basic motion sensors provide limited information about surrounding road conditions.
- Manual monitoring makes it difficult to identify faulty street lights quickly.
- Large-scale urban lighting requires continuous monitoring and maintenance.
- Conventional systems provide limited prediction and energy analytics.

Motivation

- Reduce unnecessary electricity consumption.
- Automatically adapt lighting intensity to road activity.
- Improve infrastructure monitoring and maintenance.
- Detect vehicle and pedestrian activity in real time.
- Enable faster identification of lighting faults.
- Provide centralized smart-city management and analytics.

Introduction

Why the Project Is Needed

- To reduce unnecessary electricity consumption by adjusting street-light intensity according to actual road activity.
- To provide adaptive illumination based on real-time vehicle and pedestrian movement.
- To minimize manual monitoring and control of large-scale street-light infrastructure.
- To identify faulty or abnormal street lights and support faster maintenance.
- To use Artificial Intelligence and Computer Vision for intelligent, data-driven lighting decisions.
- To provide centralized monitoring and energy analytics for efficient smart-city management.

Application / Domain of the Project

Application / Domain

Primary Domain: AI + Smart Cities

- Urban Roads
- Highways
- Residential Streets
- Industrial Areas
- Parking Areas
- Smart Campuses
- Municipal Infrastructure
- Smart-City Zones

Problem Statement

Existing Problem

- Conventional street lights often operate at fixed brightness or predefined schedules.
- High illumination may continue during low-traffic periods, causing unnecessary electricity consumption.
- Timer-based and basic sensor systems cannot fully understand real-time traffic and pedestrian conditions.
- Manual monitoring of large numbers of street lights is difficult and time-consuming.
- Faulty lights may remain unnoticed, leading to delayed maintenance and reduced road visibility.

Specific Issue to Solve

- Develop an intelligent system that dynamically controls street-light intensity based on real-time road activity.
- Detect vehicles and pedestrians and use traffic density to determine appropriate illumination.
- Reduce unnecessary energy usage while maintaining adequate lighting for road safety.
- Provide centralized monitoring, energy analytics and alerts for abnormal or faulty street lights.

Existing System

Existing Methods / Systems

- Manual ON/OFF control of street lights.
- Timer-based lighting according to predefined schedules.
- LDR-based automatic lighting based on ambient light.
- PIR or motion-sensor based lighting for detecting movement.
- Fixed-intensity LED street-lighting systems.
- Basic IoT-based monitoring and control systems.

Technologies Currently Used

- Timer controllers and programmable control units
- LDR (Light Dependent Resistor) sensors
- PIR / motion sensors
- LED street-light systems
- Electrical control panels
- Basic IoT communication modules

Existing System

Limitations and Challenges

- Fixed-brightness operation can cause unnecessary electricity consumption during low-traffic periods.
- Timer-based systems cannot adapt to real-time changes in traffic and pedestrian activity.
- Basic sensors provide limited contextual information about road conditions.
- Manual monitoring and maintenance are difficult for large numbers of street lights.
- Faulty or non-functional lights may not be detected quickly.
- Conventional systems generally lack predictive lighting control and advanced energy analytics.
- Scaling basic sensor-based systems across large urban areas can increase monitoring and maintenance complexity.

Proposed System

Proposed Solution

Develop an AI-enabled street-lighting system that dynamically controls brightness using real-time road activity.

Use camera and sensor data to detect vehicles, pedestrians and surrounding conditions.

Analyze traffic density and environmental information to determine the required illumination level.

Automatically adjust LED street-light intensity through an IoT-enabled controller.

Monitor energy consumption and street-light status through a centralized dashboard.

How It Improves the Existing System

Fixed Brightness	→	Adaptive brightness based on real-time traffic and activity.
Timer-Based Control	→	AI-driven decisions using multiple real-time inputs.
Manual Monitoring	→	Centralized dashboard for automatic monitoring.
Delayed Fault Detection	→	Automated fault identification and alerts.
Limited Analytics	→	Energy and lighting data stored for analysis.

Real-Time Data
↓
AI Decision
↓
Adaptive Lighting

Proposed System

Major Features

- Automatic ON/OFF Control
- Vehicle & Pedestrian Detection
- AI-Based Lighting Decision
- Street-Light Fault Detection
- Adaptive Brightness Control
- Traffic Density Analysis
- Energy Consumption Monitoring
- Centralized Dashboard

Expected Benefits

- Reduced unnecessary electricity consumption through adaptive lighting.
- Improved illumination based on real-time road activity.
- Less manual effort for monitoring and maintenance.
- Faster identification of faulty or abnormal street lights.
- Centralized monitoring with energy and lighting analytics.
- Scalable intelligent infrastructure for smart-city applications.

Objectives

Specific Objectives

- To develop an AI-enabled intelligent street-lighting control system for smart-city applications.
- To detect vehicles and pedestrians using real-time camera or sensor data.
- To estimate traffic density and road activity for adaptive lighting decisions.
- To dynamically adjust street-light brightness according to real-time traffic and environmental conditions.
- To reduce unnecessary electricity consumption while maintaining adequate road illumination.
- To monitor street-light status and energy usage through a centralized system.
- To detect faulty or abnormal street-light conditions and generate timely alerts.
- To provide a scalable and data-driven solution for intelligent urban infrastructure.

Literature Survey

Paper	Method	Key Finding	Limitation
Perandones et al. (2014)	LED + motion & luminosity sensors; 6LoWPAN	Adaptive dimming responds to pedestrians/vehicles and ambient light.	Campus-scale evaluation; limited AI-based traffic understanding.
A traffic-aware street lighting scheme (2015)	Networked sensors with real-time traffic-aware control	Dynamically adjusts brightness for vehicles and pedestrians.	Evaluated mainly through simulation rather than broad field deployment.
Mahoor et al. (2020)	Review of smart streetlights, communication and control strategies	Identifies energy saving, remote monitoring and fault detection as key capabilities.	Review study; does not implement a single end-to-end system.
Advanced Adaptive Street Lighting Systems (2020)	IoT architecture with adaptive brightness based on road users	Supports automatic brightness adjustment and scalable networked lighting.	Deployment and sensing architecture can add infrastructure complexity.
Chen et al. (2022)	IoT sensors + intelligent decision module + PWM/DALI control	Uses traffic/presence information to compute lighting intensity and reduce energy use.	Depends on sensor data and communication/control infrastructure.
Smart Street Light Control: Review (2023)	Survey of static, sensor, computer-vision and deep-learning controls	Shows progression from rule-based control toward AI and computer vision.	AI methods can require higher computation, data and deployment resources.

Research Gap

What Has Already Been Done?

- Smart street lights use timers, LDRs, PIR sensors and IoT-based control.
- Adaptive systems adjust lighting based on motion, traffic or environmental conditions.
- Research has explored remote monitoring, energy optimization and fault detection.

What Limitations Exist?

- Many systems depend on fixed schedules or basic sensors with limited context.
- Existing solutions may not accurately understand real-time traffic and pedestrian density.
- AI-based control can require more computation, data and reliable communication infrastructure.
- A need remains for an integrated, scalable system combining AI, adaptive brightness, monitoring and fault alerts.

Research Gap

What Is Missing in Existing Research?

- Better combination of real-time traffic detection and adaptive street-light control.
- More reliable lighting decisions than fixed schedules or basic motion sensors.
- Strong integration between traffic analysis, brightness control and energy monitoring.
- Efficient monitoring and fault detection for large numbers of street lights.

How Does the Proposed Project Address the Gap?

- Uses AI-based vehicle and pedestrian detection for real-time traffic awareness.
- Estimates traffic density and dynamically adjusts street-light brightness.
- Combines camera/sensor input with intelligent decision logic for adaptive control.
- Provides centralized monitoring, energy tracking and automatic fault alerts.

Proposed Methodology

INPUT

1. Data Acquisition



- Camera captures road scenes
- Collect traffic and pedestrian activity
- Record lighting status

PROCESSING

2. Pre-processing

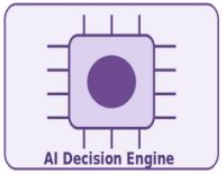
- Extract video frames
- Resize and normalize
- Noise reduction
- Prepare data for AI

AI / MODEL

3. Detection

- Detect vehicles
- Detect pedestrians
- Estimate traffic density

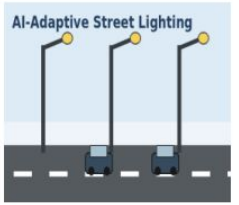
4. AI Decision



- Analyze traffic level
- Select brightness level
- Generate control decision

OUTPUT

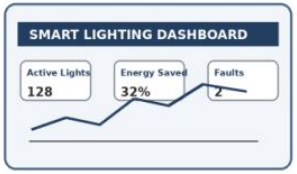
5. Control



- Send command to controller
- Adjust LED intensity
- Automatic ON/OFF

EVALUATION

6. Output



- Adaptive brightness
- Energy-efficient lighting
- Improved visibility

7. Evaluation

- Detection accuracy
- Energy consumption
- Response time
- Fault alerts

Overall Flow: Camera / Sensors → Pre-processing → AI Detection → Intelligent Decision → Adaptive Lighting → Monitoring & Evaluation

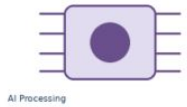
Modules

System Modules



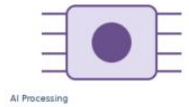
1. Data Acquisition

- Capture images/video
- Collect traffic data



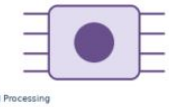
2. Image Pre-processing

- Resize & normalize
- Noise reduction



3. Vehicle & Pedestrian Detection

- Detect road users
- Estimate density



4. AI Decision Module

- Analyze traffic level
- Choose brightness



5. Adaptive Light Control

- Adjust LED intensity
- Automatic ON/OFF



6. Energy Monitoring

- Track consumption
- Measure savings



7. Fault Detection

- Identify failures
- Generate alerts



8. Dashboard & Reporting

- Live status
- History & analytics

Data → AI Analysis → Lighting Control → Energy Monitoring → Alerts → Dashboard

Technologies Used

- **Programming Language:** Python
- **Frontend / Backend:** HTML, CSS, JavaScript / Flask (Python)
- **Database:** SQLite / MySQL
- **Frameworks / Libraries:** TensorFlow, OpenCV, NumPy, Pandas
- **Cloud / Tools:** AWS, GitHub, VS Code
- **Hardware / IoT:** ESP32, LED Street Lights, LDR / PIR Sensors
- **AI/ML Algorithms:** YOLO, CNN-based Traffic Detection, Adaptive Decision Logic

Dataset / Data Source

Dataset & Training Setup

- **Dataset Source:** Public traffic and pedestrian datasets with road-scene images/videos.
- **Data Collection:** Surveillance-camera footage can be added for local road conditions and lighting scenarios.
- **Target Classes:** Vehicles, pedestrians and other relevant road users.
- **Pre-processing:** Frame extraction, resizing, normalization, noise reduction and data cleaning.
- **Data Augmentation:** Rotation, scaling, cropping, brightness variation and horizontal flipping.
- **Training Setup:** Training, validation and testing sets are prepared separately to evaluate model performance.
- **Model Training:** AI detection model learns traffic patterns and supports adaptive street-light brightness decisions.

Algorithm / Model

Model	Working Principle	Why Selected	Expected Advantage
YOLO	Single-stage object detection identifies vehicles and pedestrians from camera frames using bounding boxes and confidence scores.	Suitable for real-time road-scene detection and traffic-density estimation.	Fast inference and real-time detection for adaptive lighting.
CNN	Learns visual features from road images and classifies detected objects or traffic-related patterns.	Provides feature extraction for reliable image-based recognition.	Improved classification and better handling of visual variations.
OpenCV	Processes video frames through resizing, filtering, enhancement and image preparation before AI inference.	Lightweight and integrates easily with Python-based AI pipelines.	Efficient preprocessing and smoother real-time processing.
Decision Logic	Combines traffic density, time and environmental conditions to determine the required light intensity.	Directly converts AI observations into lighting-control actions.	Adaptive brightness, reduced energy use and improved road visibility.

Expected Outcome

- **Intelligent Lighting:** Automatically adjust street-light brightness according to real-time traffic and pedestrian activity.
- **Energy Efficiency:** Reduce unnecessary electricity consumption by avoiding constant full-intensity operation.
- **Improved Safety:** Maintain adequate illumination when vehicles or pedestrians are detected on the road.
- **Real-Time Monitoring:** Provide centralized monitoring of street-light status, activity and energy usage.
- **Fault Detection:** Identify lighting failures and generate alerts for faster maintenance.
- **Reduced Manual Effort:** Automate lighting decisions and reduce the need for continuous manual monitoring.
- **Scalable Smart-City Solution:** Support deployment across roads, campuses and other urban infrastructure.

Performance Evaluation

Evaluation Metrics – Project Values

AI/ML Performance

- Detection Accuracy – Measures correct vehicle and pedestrian detection.
- Precision & Recall – Evaluates false detections and missed road users.
- F1-Score – Measures the balance between precision and recall.
- Inference Speed – Measures real-time processing performance (FPS).

Software/System Performance

- Response Time – Time taken to convert traffic analysis into a lighting decision.
- Energy Consumption – Compare adaptive operation with constant full-intensity lighting.
- Lighting Control Accuracy – Verify that brightness changes match detected traffic conditions.
- Fault Detection – Measure identification of failed or abnormal street lights.

Work Completed

- **Problem Identification:** Need for accurate, real-time street-light control to reduce unnecessary energy consumption and maintain road safety.
- **Literature Survey:** Studied conventional street lighting, timer-based systems, LDR/PIR sensor control, IoT-based lighting and AI-driven adaptive lighting.
- **Research Gap:** Existing systems often depend on fixed schedules or basic sensors and have limited understanding of real-time traffic and pedestrian conditions.
- **Objectives:** Detect vehicles and pedestrians, estimate traffic density, adjust light intensity automatically, reduce energy usage and monitor street-light status.
- **Proposed Methodology:** Camera/Sensor Input → Image Pre-processing → Vehicle & Pedestrian Detection → Traffic Density Analysis → AI Decision → Adaptive Light Control.
- **Architecture/Design:** Input images/video → Pre-processing → AI-based detection → Traffic analysis → Lighting decision → LED control → Monitoring and alerts.
- **Technology Selection:** Python, Flask, OpenCV, TensorFlow, YOLO, HTML, CSS, JavaScript, ESP32 and LED street-light control.
- **Dataset Identification:** Road-scene/traffic datasets containing vehicle and pedestrian classes, with camera-based data considered for real-world testing.

Future Work

- Integrate more advanced computer-vision models for improved vehicle and pedestrian detection.
- Deploy the system on edge devices for real-time processing with lower latency.
- Add weather, ambient-light and road-condition information to improve lighting decisions.
- Develop predictive maintenance using historical street-light fault and energy data.
- Expand the system to large-scale smart-city deployments with centralized IoT management.
- Add mobile/web notifications and detailed energy-consumption analytics for administrators.

References

1. Perandones, J. M. et al. (2014). Smart Street Lighting Using LED and Motion/Luminosity Sensors. 6LoWPAN-based adaptive lighting system.

2. A Traffic-Aware Street Lighting Scheme for Smart Cities Using Autonomous Networked Sensors (2015). Traffic-aware adaptive street-light control.

3. Mahoor, M. et al. (2020). State-of-the-Art in Smart Streetlight Systems: A Review. Smart streetlight communication and control strategies.

4. Advanced Adaptive Street Lighting Systems for Smart Cities (2020). IoT-based adaptive street-lighting architecture.

5. Chen, X. et al. (2022). IoT-Based Smart and Intelligent Smart City Energy Optimization. Intelligent energy and lighting control using IoT.

6. Smart Street Light Control: A Review on Methods, Innovations, and Extended Applications (2023). Review of sensor-, computer-vision- and AI-based street-light control.

INTELLIGENT STREET LIGHTING CONTROL SYSTEM

PROJECT REPORT

Project Work – Phase I

Student Name: Pranav Balaji. P

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ABSTRACT

Street lighting is an important component of urban infrastructure because it supports visibility, road safety, and pedestrian movement during low-light conditions. Conventional street-lighting systems often use fixed brightness levels, timers, or simple sensors. These methods may continue operating at high illumination when road activity is low and may provide limited information about traffic conditions or faulty lights.

The Intelligent Street Lighting Control System proposes an AI-enabled and IoT-supported approach to adaptive lighting. Camera or sensor inputs are processed to identify vehicles and pedestrians, estimate traffic density, and determine a suitable illumination level. An IoT-enabled controller then adjusts LED street-light intensity. A centralized dashboard is proposed for monitoring light status, energy use, alerts, and historical information.

The proposed workflow combines data acquisition, image pre-processing, computer-vision detection, intelligent decision logic, adaptive light control, and performance evaluation. Python, Flask, OpenCV, TensorFlow, YOLO, HTML, CSS, JavaScript, a database, ESP32, and LED lighting are identified as potential technologies. The project is intended to reduce unnecessary electricity consumption while maintaining adequate illumination and supporting quicker fault identification. Quantitative performance results are not claimed in this report because the supplied project presentation describes the proposed design and expected outcomes rather than completed deployment measurements.

CHAPTER 1: INTRODUCTION

1.1 Background

Street lighting helps drivers, cyclists, and pedestrians navigate roads safely at night. In many conventional installations, lights operate according to a fixed schedule or remain at a constant brightness. Although simple to deploy, these approaches do not always reflect actual road use. A quiet road may receive the same illumination as a busy road, resulting in energy being used when full brightness may not be necessary.

Smart-city infrastructure increasingly requires systems that can respond to changing conditions and provide centralized monitoring. Artificial Intelligence (AI), computer vision, and the Internet of Things (IoT) can be combined to analyse road activity and control lighting more intelligently.

1.2 Project Overview

The Intelligent Street Lighting Control System is designed to detect road users, estimate traffic density, and adjust LED brightness according to observed activity and relevant environmental inputs. The system also aims to monitor lighting status and energy consumption, identify abnormal conditions, and provide alerts through a centralized interface.

1.3 Motivation

The project is motivated by the need to reduce unnecessary electricity use, decrease dependence on manual monitoring, respond to real-time vehicle and pedestrian movement, and identify lighting failures more quickly. It also explores how data-driven control can support scalable municipal and smart-campus infrastructure.

1.4 Application Areas

Potential application areas include urban roads, highways, residential streets, industrial areas, parking areas, smart campuses, municipal infrastructure, and designated smart-city zones.

CHAPTER 2: PROBLEM DEFINITION AND OBJECTIVES

2.1 Existing Problem

Conventional systems commonly rely on manual ON/OFF operation, timers, ambient-light sensors, motion sensors, or fixed-intensity LED lighting. Timers do not understand current traffic density. Basic motion sensors may detect movement without providing a broader understanding of road activity. Manual supervision of a large number of lights is time-consuming, and faults may remain unnoticed until a person reports them.

2.2 Problem Statement

The project addresses the need for a street-lighting system that can dynamically determine illumination from real-time road activity, reduce unnecessary energy use without compromising adequate road visibility, and support centralized monitoring and fault alerts.

2.3 Aim

To design an AI-enabled intelligent street-lighting control system that adapts LED brightness to detected traffic and pedestrian activity and provides monitoring of energy usage and light status.

2.4 Specific Objectives

- Detect vehicles and pedestrians using camera or sensor data.
- Estimate traffic density and road activity for lighting decisions.
- Adjust street-light brightness dynamically according to traffic and environmental conditions.
- Reduce unnecessary electricity consumption while maintaining suitable road illumination.
- Monitor street-light status and energy usage through a centralized system.
- Identify faulty or abnormal street-light conditions and generate alerts.
- Develop a scalable, data-driven approach for smart-city applications.

2.5 Scope

The scope covers the proposed software pipeline, AI-based road-user detection, lighting decision logic, IoT-enabled control, energy monitoring, fault alerts, and a dashboard. Actual energy savings, detection accuracy, and response time must be established through implementation and testing; they are not assumed in advance.

CHAPTER 3: EXISTING SYSTEM AND LIMITATIONS

3.1 Existing Methods

- Manual ON/OFF control operated by maintenance personnel.
- Timer-based control using predefined schedules.
- LDR-based automatic operation based on ambient brightness.
- PIR or other motion-sensor-based lighting.
- Fixed-intensity LED street-lighting systems.
- Basic IoT monitoring and control solutions.

3.2 Technologies in Existing Systems

Common components include timer controllers, programmable control units, LDR sensors, PIR/motion sensors, LED lighting, electrical control panels, and basic IoT communication modules.

3.3 Limitations

- Fixed brightness may consume unnecessary power during low-traffic periods.
- Timer-based systems cannot respond directly to real-time changes in traffic.
- Basic sensors provide limited contextual understanding of road conditions.
- Manual monitoring becomes difficult as the number of lights increases.
- Faulty or non-functional lights may not be detected quickly.
- Conventional systems may lack predictive control and detailed energy analytics.

- Large-scale deployment can increase communication, monitoring, and maintenance complexity.

CHAPTER 4: PROPOSED SYSTEM

4.1 Proposed Solution

The proposed system uses camera and/or sensor inputs to observe road activity. Video frames are prepared for analysis, and an AI-based detection stage identifies vehicles and pedestrians. The system estimates traffic density and uses intelligent decision logic to select an appropriate brightness level. An IoT-enabled controller applies the command to the LED street light. A centralized dashboard presents light status, energy information, and alerts.

4.2 Major Features

- Automatic ON/OFF control.
- Adaptive brightness control.
- Vehicle and pedestrian detection.
- Traffic-density analysis.
- AI-based lighting decisions.
- Energy-consumption monitoring.
- Street-light fault detection and alerts.
- Centralized dashboard, history, and reporting.

4.3 Expected Benefits

Adaptive illumination is expected to reduce unnecessary electricity use compared with constant full-intensity operation. Traffic-aware control can help provide more illumination when road users are present. Centralized status information can reduce manual inspection effort, while automated alerts can help maintenance teams respond to abnormal lights. These benefits require validation through prototype and field testing.

4.4 High-Level Workflow

Camera/Sensors → Image Pre-processing → Vehicle and Pedestrian Detection → Traffic-Density Analysis → AI Decision → Adaptive Light Control → Energy Monitoring and Alerts → Dashboard.

CHAPTER 5: LITERATURE SURVEY AND RESEARCH GAP

5.1 Literature Survey Summary

The supplied presentation identifies several representative research directions: LED lighting with motion and luminosity sensors; traffic-aware control using networked sensors; reviews of smart-streetlight communication and control; IoT-based adaptive lighting architectures; intelligent decision modules using traffic or presence information; and surveys of sensor-, computer-vision-, and deep-learning-based control.

Literature summary

Reference / topic	Approach described	Key point	Limitation noted in the presentation
Perandones et al. (2014)	LED, motion and luminosity sensors; 6LoWPAN	Adaptive dimming based on road users and ambient light	Campus-scale evaluation; limited AI traffic understanding
Traffic-aware lighting scheme (2015)	Networked sensors and real-time traffic-aware control	Brightness adapts to vehicles and pedestrians	Mainly simulation-based evaluation
Mahoor et al. (2020)	Review of smart streetlights and control strategies	Energy saving, remote monitoring and fault detection	Review rather than a single end-to-end implementation
Advanced adaptive lighting (2020)	IoT architecture with adaptive brightness	Automatic brightness and networked lighting	Deployment and sensing add complexity
Chen et al. (2022)	IoT sensors, decision module, PWM/DALI control	Traffic/presence information informs lighting intensity	Depends on sensing and communication infrastructure
Smart street-light control review (2023)	Survey of static, sensor, vision and deep-learning methods	Shows progression toward AI and computer vision	AI may require more computation, data and resources

5.2 Research Gap

The presentation identifies a need to integrate real-time traffic detection and adaptive lighting more tightly. Existing solutions may depend on fixed schedules or basic sensors, may not estimate traffic and pedestrian density accurately, and may not combine lighting decisions, energy monitoring, and fault alerts in one scalable workflow. AI-based systems also need to account for computation, training data, and communication reliability.

5.3 How the Proposed Project Addresses the Gap

- Use AI-based vehicle and pedestrian detection for traffic awareness.
- Estimate traffic density and adjust brightness dynamically.
- Combine camera/sensor input with decision logic and lighting control.
- Provide centralized monitoring, energy tracking, and automatic fault alerts.

The references above are transcribed and summarized from the supplied presentation. Full bibliographic metadata should be verified against the original publications before a final academic submission.

CHAPTER 6: METHODOLOGY AND SYSTEM DESIGN

6.1 Data Acquisition

The input stage captures road-scene images or video and may collect sensor readings and street-light status. Public traffic and pedestrian datasets are proposed for initial model development, with local surveillance-camera footage considered for testing under relevant road and lighting conditions.

6.2 Image Pre-processing

Video is divided into frames. Frames may be resized and normalized, and noise reduction or other image preparation can be applied. Data cleaning and consistent input preparation help make the detection pipeline more reliable.

6.3 Vehicle and Pedestrian Detection

The proposed AI stage detects vehicles, pedestrians, and other relevant road users. YOLO is identified as a candidate real-time object-detection approach. Detection results may include object classes, bounding boxes, and confidence scores.

6.4 Traffic-Density Estimation

The detected road users are analysed to estimate traffic activity. The system can use the estimated density, time, and environmental conditions as inputs to its lighting decision. Exact thresholds and decision rules must be defined and validated during implementation.

6.5 Intelligent Decision and Lighting Control

Decision logic maps traffic activity and other inputs to a target brightness level. The controller then adjusts LED intensity or switches the light ON/OFF as appropriate. The final implementation must use a safe, electrically suitable driver/controller and preserve minimum illumination requirements.

6.6 Energy Monitoring and Fault Alerts

The system is intended to track energy use and lighting status. Abnormal states or suspected failures can be flagged for maintenance. Energy savings should be calculated by comparing measured consumption under adaptive operation with a defined baseline, using equivalent time periods and conditions.

6.7 Dashboard and Reporting

A web dashboard can display current status, traffic activity, selected brightness, energy records, fault alerts, and historical trends. The supplied presentation identifies HTML, CSS, JavaScript, Flask, and SQLite/MySQL as possible implementation technologies.

6.8 Proposed Evaluation

Evaluation should cover object-detection accuracy, precision, recall, F1-score, inference speed in frames per second, end-to-end response time, lighting-control correctness, energy consumption, and fault-detection performance. Tests should include different traffic levels, lighting conditions, and relevant environmental variations.

CHAPTER 7: SYSTEM MODULES

7.1 Data Acquisition Module

Captures images/video, gathers traffic information, and records lighting status for downstream processing.

7.2 Image Pre-processing Module

Resizes and normalizes frames and applies suitable noise reduction or image preparation before inference.

7.3 Vehicle and Pedestrian Detection Module

Identifies road users in frames and provides detections that can be used to estimate traffic density.

7.4 AI Decision Module

Combines traffic level with time and environmental inputs to determine the target brightness or control action.

7.5 Adaptive Light Control Module

Sends the selected control command to the IoT-enabled controller and adjusts LED intensity or ON/OFF state.

7.6 Energy Monitoring Module

Stores or displays energy-consumption information and supports comparison between adaptive and baseline operation.

7.7 Fault Detection Module

Identifies possible failed or abnormal lights from available status information and generates alerts for maintenance.

7.8 Dashboard and Reporting Module

Provides a centralized view of current status, alerts, energy information, and historical records.

CHAPTER 8: TECHNOLOGIES AND DATASET

8.1 Proposed Technology Stack

Technology stack

Category	Technology identified	Proposed role
Programming language	Python	AI pipeline and backend logic
Frontend / backend	HTML, CSS, JavaScript / Flask	Dashboard and web application
Database	SQLite or MySQL	Store status, records, and alerts
Libraries	TensorFlow, OpenCV, NumPy, Pandas	Model support, image processing, and data handling
AI model	YOLO; CNN-based detection	Vehicle and pedestrian detection
Hardware / IoT	ESP32, LED street lights, LDR/PIR sensors	Controller and optional sensor inputs
Tools / cloud	AWS, GitHub, VS Code	Development, version control, and possible deployment

8.2 Dataset and Training Setup

The presentation proposes public road-scene datasets containing vehicle and pedestrian examples. Additional local camera footage may be collected for road-specific testing. Frames are pre-processed, and data augmentation may include rotation, scaling, cropping, brightness variation, and horizontal flipping. Separate training, validation, and testing sets should be used to evaluate generalization.

8.3 Model Choices

YOLO is proposed for single-stage object detection suited to real-time road scenes. CNN-based techniques can learn visual features for recognition and classification. OpenCV supports frame extraction, resizing, filtering, and image preparation. Decision logic converts the resulting traffic information into a lighting-control action.

8.4 Implementation Considerations

- Select a dataset whose licence permits the intended academic use.
- Avoid data leakage between training, validation, and test sets.
- Evaluate performance during day, night, low-light, and varying traffic conditions.
- Measure inference speed on the actual target hardware before selecting deployment settings.
- Protect camera footage and avoid retaining personally identifiable imagery unnecessarily.

CHAPTER 9: PERFORMANCE EVALUATION

9.1 AI/ML Metrics

Detection accuracy measures the correctness of detections. Precision indicates how many predicted detections are correct, while recall indicates how many relevant road users are detected. The F1-score balances precision and recall. Inference speed, commonly measured in frames per second, indicates whether processing can keep pace with incoming video.

9.2 System Metrics

Response time measures the delay from analysing traffic to issuing a lighting command. Energy consumption should be measured over comparable conditions and compared with constant full-intensity operation. Lighting-control accuracy checks whether the commanded brightness matches the defined decision policy. Fault-detection evaluation should measure correctly identified faults and missed or false alerts.

9.3 Testing Plan

- Test low, medium, and high traffic-density scenarios.
- Test vehicle-only, pedestrian-only, and mixed road-user scenes.
- Test different illumination and weather conditions where data is available.
- Compare adaptive control against a clearly documented baseline.
- Record detection metrics, response time, energy use, and fault alerts.
- Report measured results with test conditions; do not claim savings or accuracy without evidence.

9.4 Results Status

The source presentation lists evaluation metrics but does not provide measured numerical results. Accordingly, this report describes the evaluation framework and does not invent accuracy, response-time, or energy-saving values.

CHAPTER 10: WORK COMPLETED AND FUTURE ENHANCEMENTS

10.1 Work Completed / Defined in Phase I

- Identified the problem of inefficient fixed-brightness lighting and delayed fault detection.
- Reviewed conventional timer, LDR/PIR, IoT, and AI-driven lighting approaches.
- Defined the research gap and project objectives.
- Prepared a proposed processing workflow and system architecture.
- Selected candidate technologies, including Python, Flask, OpenCV, TensorFlow, YOLO, ESP32, and LED control.
- Identified public traffic datasets and a preliminary training approach.

These items represent the planning and design work described in the supplied Phase I presentation. They should not be interpreted as confirmation that a complete prototype or field deployment has already been implemented.

10.2 Future Work

- Integrate and evaluate advanced computer-vision models for more reliable detection.
- Deploy the model on edge devices to reduce processing latency.
- Incorporate weather, ambient-light, and road-condition inputs.
- Develop predictive maintenance using historical fault and energy data.
- Extend centralized IoT management for larger deployments.
- Add mobile/web notifications and detailed energy analytics.
- Conduct controlled prototype and field tests and publish measured results.

CHAPTER 11: CONCLUSION

The Intelligent Street Lighting Control System proposes an integrated approach to street-light management by combining computer vision, AI-based traffic analysis, adaptive LED control, IoT connectivity, energy monitoring, and centralized fault reporting. Compared with fixed schedules and simple motion-based systems, the proposed design aims to make lighting decisions more responsive to vehicle and pedestrian activity.

The project's main contribution at the design stage is the definition of an end-to-end workflow that connects road-scene input to detection, traffic-density estimation, brightness selection, controller action, and monitoring. Python, Flask, OpenCV, TensorFlow, YOLO, ESP32, and a database form the proposed technology stack. Future implementation should validate detection quality, response time, safe illumination, fault identification, and energy consumption under representative conditions.

The system has potential for urban roads, residential streets, parking areas, industrial zones, and smart campuses. Its practical value will depend on reliable sensing, appropriate lighting thresholds, secure communication, electrical safety, and measured evidence from testing. No numerical performance claims are made here because the supplied presentation does not include completed experimental results.

REFERENCES

The following references are listed in the supplied presentation. Verify the full publication details, venue, page numbers, and DOI/URL from the original publications before final submission.

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